Technical Requirements — Temperature-controlled Optical Readout Module

Echo Yeast Hardware Module (current revision)

Revision: 14 October 2026 Prepared by: Echo Yeast Hardware Team

1 Scope

This specification defines the terms, technical requirements, test methods, inspection rules, and marking, packaging, transport and storage requirements for the hardware module of the Echo Yeast whole-cell biosensor.

The module holds and incubates a liquid sample containing the yeast sensor (cuvette or our in-house collection cup), provides a 485 nm excitation path and a 528 nm fluorescence collection path, and outputs raw measurement data for recording. It is intended for laboratory validation and small-batch prototypes.

2 Normative References

- GB/T 38517-2020 Particles — General rules for bioaerosol sampling and analysis
- GB/T 1804-m General tolerances — Tolerances for linear and angular dimensions without individual tolerance indications
- GB 9706.1 Medical electrical equipment — Part 1: General requirements for basic safety
- ISO 10993 Biological evaluation of medical devices
- WS 648-2019 General hygienic requirements for air disinfection machines (applicable when used as an add-on module)

3 Terms and Definitions

Terms defined in GB/T 38517-2020 apply. In this document, "module" refers to the unit formed by the heated block, the detachable optical plugs, the electronics and the power supply.

4 Technical Requirements

4.1 Overall Performance

Item	Requirement
Purpose	Temperature-controlled incubation and fluorescence readout of the enriched liquid sample from airborne influenza A virus aerosols; provides a stable temperature environment and an optical readout path for the yeast whole-cell biosensor
Sample vessel	Test stage: standard 10 mm path-length cuvette. Project stage: in-house 3D-printed collection cup
Working volume	Cuvette type ≥ 1.5 mL; collection-cup type 20 mL liquid medium (with yeast)
Set point	30 °C (yeast culture temperature)
Temperature accuracy	±0.5 °C (steady state, within 10 min)
Temperature uniformity	Top-to-bottom difference of the liquid in the vessel ≤ 0.5 °C
Heat-up time	Room temperature (25 °C) to 30 °C in ≤ 5 min
Excitation wavelength	485 ± 5 nm
Emission wavelength	528 ± 10 nm
Optical channel	Single channel; excitation and collection axes at 90°

Item	Requirement
Target	HA protein on the surface of influenza A virus (H1N1), driving yEGFP expression through the GPCR pathway
Detection type	Qualitative (present / absent); signal expressed as relative fluorescence intensity
Readout time	≤ 10 s per reading (including settling delay); incubation and induction follow the experimental protocol
Power supply	DC 24 V / 2 A adapter; stepped down internally to 12 V and 5 V
Power consumption	≤ 30 W peak during heating; ≤ 10 W steady state
Module dimensions	Heated block plus optical plugs ≤ 120 × 90 × 80 mm (excluding external power supply and cabling)
Weight	≤ 1.5 kg for the module itself (excluding the adapter)

4.2 Structural Requirements

4.2.1 Heated Block and Vessel Interface

Item	Requirement
Print material	PLA (white / black); inner surfaces painted matte black or lined with light-absorbing fabric to suppress stray light
Print process	FDM, layer height 0.1–0.2 mm, wall thickness ≥ 2 mm, infill ≥ 20%
Dimensional accuracy	Mating surfaces ±0.2 mm; general outline ±0.5 mm
Heated block body	Rectangular block with two Ø vessel bores (cuvette / collection cup), bore depth ≥ 30 mm; optical channels machined on the side faces
Optical channel	8 mm bore; channel axis perpendicular to and intersecting the vessel bore axis; coaxiality of the two ends ≤ 0.3 mm
Vessel fit	Clearance to the vessel wall ≤ 0.3 mm; filled with thermal pad or a thin sleeve where necessary
Detachable plugs	Emitter_Plug, Receiver_Plug and Heater_Plug are independent plug-in parts; no play once inserted; removal must not affect the block
Plug locating	Cylindrical surface plus shoulder; no play after 20 insert / remove cycles
Light sealing	Optical channel and plug-to-block joints must be fully light-tight; plugs wrapped in ≥ 12 mm black heat-shrink tubing, gaps sealed with black silicone or hot glue
Light-tightness	With the excitation source off under room lighting, receiver output drift ≤ 1% of full scale
Heat dissipation	Aluminium heat sink (70 × 36 × 100 mm) and 40 × 40 × 10 mm, 12 V quiet fan mounted under the block
Assembly	Plugs and vessel can be inserted and removed without tools; M3 self-tapping holes or heat-set inserts
General tolerance	Per GB/T 1804-m; unspecified fillets R0.5, unspecified chamfers C0.5

4.2.2 Detachable Optical Plugs

Emitter_Plug, Receiver_Plug and Heater_Plug are independent plug-in parts. Emitter_Plug carries the 485 nm LED, the focusing lens and the excitation filter; Receiver_Plug carries the emission filter, the focusing lens and the BPW34 photodiode; Heater_Plug routes the heating-film leads and the PT100 into the block. The plug-to-block interface is identical across revisions, so switching vessels (cuvette ↔ in-house collection cup) requires no redesign of the plugs.

4.3 Temperature Control

Item	Requirement
Heating element	PI polyimide heating film, 60 × 70 mm, 12 V / 25 W, two pieces

Item	Requirement
Mounting	Bonded with PI-based high-temperature thermal tape (0.1 mm); edges fixed and sealed with K-5204K thermal adhesive
Sensor	PT100 platinum resistor (Class A, 3-wire), inserted into a blind bore filled with thermal grease
Transmitter	MAX31865 module, SPI, 16-bit ADC, resolution ≤ 0.03 °C
Control range	Room temperature to 40 °C
Set point	30 °C
Accuracy	±0.5 °C (steady state)
Power drive	IRF520 MOSFET driver module, PWM with PID algorithm
Over-temperature (hardware)	KSD9700 thermal fuse (60 °C, 10 A, metal case, normally closed) in series with the heater for physical cut-off
Over-temperature (software)	Software threshold ≤ 45 °C; PWM output disabled immediately when exceeded
Cooling	40 mm, 12 V quiet fan plus aluminium heat sink
Thermal interface	6 W/(m·K) thermal pad (200 × 400 × 0.3 mm, cut to size) and K-5204K thermal adhesive
Insulation	Insulation resistance between heater leads and metal body ≥ 2 MΩ

4.4 Optical Detection

Item	Requirement
Excitation source	485 nm high-brightness LED with adjustable constant-current driver (350 / 500 mA)
Excitation focusing	Focusing lens in the emitter end converging the LED output to the centre of the vessel; spot diameter ≤ 3 mm
Excitation filter	Ø10 mm narrow-band filter, CWL 485 nm, FWHM 30 nm, OD3
Emission filter	Ø10 mm narrow-band filter, CWL 528 nm, FWHM 30 nm, OD3
Emission focusing	Focusing lens in the receiver end converging fluorescence onto the photodiode active area
Photodetector	BPW34 silicon PIN photodiode (through-hole)
Geometry	Excitation and collection axes at 90°, both perpendicular to and intersecting the vessel bore axis at the same height
Transimpedance stage	OPA350 single-channel op-amp as transimpedance amplifier; feedback resistor selected to match signal level, gain adjustable
Light sealing	Emitter and receiver ends wrapped in black heat-shrink tubing; joints to the block sealed against light
Alignment	After insertion, the intersection of the optical axes deviates from the vessel bore axis by ≤ 0.5 mm
Serviceability	LED, filters, focusing lenses and photodiode are detachable and can be replaced or cleaned individually
Detection limit	Against a blank medium without HA as negative control, target signal-to-noise ratio S/N ≥ 3; the actual limit is determined experimentally

4.5 Signal Acquisition and Temperature-control Unit

Item	Requirement
Core board	Arduino Nano development board; handles sequencing, ADC readout and the temperature-control algorithm
ADC	ADS1115 high-precision ADC, 16-bit, 4 differential channels, I²C, programmable gain
Temperature readout	PT100 resistance read via SPI from the MAX31865 and converted to temperature
Power output	PWM output to the IRF520 driver module to set heater power

Item	Requirement
Acquisition	Switches the 485 nm LED on and off on a fixed timing sequence, reads the receiver voltage and logs background and signal continuously
Data output	Timestamp, temperature and raw ADC counts emitted over USB / serial in a fixed format for offline analysis and archiving
Resolution	Effective number of bits ≥ 14; background noise ≤ 2 LSB
Sampling rate	Temperature ≥ 1 Hz; optical signal ≥ 10 Hz (adjustable)
Time base	On-board crystal oscillator; drift ≤ 1 s over a continuous run of ≥ 1 h
Power distribution	24 V input stepped down by LM2596 to 12 V (heater, fan) and 5 V (logic)

4.6 Power Supply and Auxiliaries

Item	Requirement
Input	DC 24 V / 2 A adapter (48 W) with 5.5 × 2.1 mm barrel connector
Step-down	LM2596S adjustable DC-DC module, 24 V to 12 V and 5 V
Filtering	100 µF / 16 V electrolytic capacitors placed close to each supply node
Circuit carrier	12 × 18 cm single-sided perfboard for soldered assembly; breadboard for the debugging stage
Power wiring	22 AWG silicone wire (red / black), rated ≥ 105 °C, for heater and supply loops
Signal wiring	30 AWG extra-flexible silicone wire for wiring inside the plugs and between boards
Board-to-board	21 cm Dupont jumpers (male-male, male-female, female-female)
SMD adapters	SOT89 and SOT223 adapter boards for through-hole assembly of SMD parts
Passives	1/4 W metal-film resistor kit (25 common values, 500 pieces); 5 mm red / green / yellow indicator LEDs
Terminals	KF301-2P / 3P, 5.08 mm pitch, interlocking
Soldering tools	936 constant-temperature soldering station (60 W), 0.8 mm rosin-cored lead-free solder, desoldering wick
Measurement	UNI-T UT33A auto-ranging digital multimeter
Fixing and sealing	40 W hot glue gun and glue sticks for securing parts and sealing light paths
Routing	Power and signal wiring routed separately to keep switching noise away from the microampere-level photocurrent signal

4.7 Safety and Biosafety

Item	Requirement
Electrical safety	In line with the relevant clauses of GB 9706.1: metal parts grounded, heater and signal circuits routed separately
Over-temperature	KSD9700 hardware fuse and software threshold as dual protection
Over-current	Fuse or resettable fuse on the supply side; current-limiting element in the heater loop
Surface temperature	Touchable surface ≤ 45 °C; insulating shroud over the heated face
Biosafety	Any work with live virus must be carried out in a laboratory of the appropriate biosafety level with suitable PPE, per GB/T 38517-2020 clause 4.5.4
Sample handling	Collected liquid samples are stored and transported per GB/T 38517-2020 clause 4.6.7 (0–10 °C, delivered within 10 h)
Disinfection	Sampling head, vessel and operators' hands disinfected before and after sampling, per GB/T 38517-2020 clause 4.5.5
Waste	Waste liquid and disposables containing biological samples are handled as infectious medical waste and never discharged to drains
Sterilisation compatibility	Vessel and plugs tolerate 75% ethanol wiping; autoclaving requires prior verification that PLA parts do not deform

4.8 Firmware Requirements

- PID temperature control: PWM duty adjusted from the PT100 reading to hold the vessel bore at 30 ± 0.5 °C;
- Optical acquisition: the 485 nm LED is switched on according to a fixed timing sequence, the photodiode voltage is read after a settling delay and converted into a relative fluorescence intensity;
- Sequence logging: timestamp, temperature and raw ADC counts written out in a fixed format for offline analysis and archiving;
- Protection logic: heating output is disabled and an error flag set immediately on over-temperature, sensor open circuit or short circuit.

5 Test Methods

Item	Method
Temperature accuracy	Set 30 °C; after 20 min stabilisation, measure the vessel bore with a calibrated thermometer (or a second PT100); deviation ≤ ±0.5 °C
Temperature uniformity	Measure at several heights inside the vessel bore; difference ≤ 0.5 °C
Heat-up time	Record the time from room temperature to 30 °C; ≤ 5 min
Over-temperature protection	Break the temperature feedback so heating runs continuously; confirm the KSD9700 cuts the heater circuit before 60 °C
Optical alignment	Insert and remove the plugs 10 times; check spot position repeatability with a paper screen or visible light; deviation ≤ 0.5 mm
Light-tightness	With the excitation source off under room lighting, acquire for 5 min; output drift ≤ 1% of full scale
Background and noise	Fill with blank medium without yeast, acquire continuously; compute mean and standard deviation of the background
Fluorescence response	Load HA-induced and uninduced yeast samples; compare readings on the 528 nm channel
Repeatability	Measure the same positive sample ≥ 5 times; relative standard deviation RSD ≤ 5% (target, to be confirmed by data)
Re-assembly consistency	Disassemble and reassemble the vessel and plugs 5 times; reading difference ≤ 5%

6 Inspection Rules

- First-article inspection: dimensional report, vessel-to-block fit record and optical alignment record for the first unit of each batch;
- Routine inspection: temperature accuracy, light-tight background, optical alignment and insulation resistance for every module;
- Type inspection: full inspection whenever a new design is finalised or the structure, optical layout or heating scheme changes.

7 Marking, Packaging, Transport and Storage

- Marking: model, name, supply rating, manufacturer and date on the module; function and insertion direction marked on each plug;
- Packaging: protective film on optical surfaces, plugs packed separately, dust-, scratch- and static-proof; inspection record enclosed;
- Transport: avoid high temperature, humidity and strong vibration;
- Storage: 0–40 °C, relative humidity ≤ 80%, no corrosive gases.

8 Notes

- This module is the readout unit paired with the aerosol sampling front end, and can also be used on its own for laboratory validation of the yeast sensor.
- This revision covers the three core functions of temperature control, optics and signal acquisition; the user interface and host software lie outside its scope and will be specified separately in a later stage.
- All unspecified tolerances per GB/T 1804-m; unspecified fillets R0.5, unspecified chamfers C0.5.
- The optical channel bore, filter centre wavelengths and focusing-lens focal lengths are the key parameters of this revision; any change requires the light-tightness and alignment tests to be repeated.
- The "working volume" and "response time" clauses reflect the latest experimental data: working volume is ≥ 1.5 mL for the cuvette type and 20 mL liquid medium for the collection-cup type; under ideal laboratory conditions a 528 nm fluorescence signal is detectable 5–15 min after the sample is added, and the response time of the complete module is to be verified experimentally.